

```
-0.6924953
gear 0.4802848 -0.4926866 -0.5555692 -0.1257043 0.69961013
-0.5832870
carb -0.5509251 0.5269883 0.3949769 0.7498125 -0.09078980
0.4276059

      qsec      vs      am      gear      carb
mpg  0.41868403 0.6640389 0.59983243 0.4802848 -0.55092507
cyl -0.59124207 -0.8108118 -0.52260705 -0.4926866 0.52698829
disp -0.43369788 -0.7104159 -0.59122704 -0.5555692 0.39497686
hp -0.70822339 -0.7230967 -0.24320426 -0.1257043 0.74981247
drat 0.09120476 0.4402785 0.71271113 0.6996101 -0.09078980
wt -0.17471588 -0.5549157 -0.69249526 -0.5832870 0.42760594
qsec 1.00000000 0.7445354 -0.22986086 -0.2126822 -0.65624923
vs 0.74453544 1.0000000 0.16834512 0.2060233 -0.56960714
am -0.22986086 0.1683451 1.00000000 0.7940588 0.05753435
gear -0.21268223 0.2060233 0.79405876 1.0000000 0.27407284
carb -0.65624923 -0.5696071 0.05753435 0.2740728 1.00000000
```

The *corrplot()* function can be used to display a correlation matrix. You can install the same in an R session using the following command:

```
> install.packages("corrplot")
```

```
Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.1'
```

```
...
** testing if installed package can be loaded from temporary location
```

```
** testing if installed package can be loaded from final location
```

```
** testing if installed package keeps a record of temporary installation path
```

```
* DONE (corrplot)
```

After loading the *corrplot* library, we can view the plot for the *mtcars* data set as follows:

```
> library("corrplot")
```

```
corrplot 0.92 loaded
```

```
> corrplot(cor(mtcars), method = "circle")
```

You can also restrict the plot to the upper segment by using the 'type' argument. For example:

```
> corrplot(cor(mtcars), method = "number", type = "upper")
```

The *corrplot()* function accepts the following arguments:

Argument	Description
corr	the correlation matrix
method	'circle', 'square', 'ellipse', 'number', 'pie', 'shade' and 'colour'
type	'full', 'upper', or 'lower'
col	specifies a vector colour of glyphs
bg	background colour
title	title of the graph
add	logical value to add plot to an existing graph
diag	logical value to display the correlation coefficients
order	'original', 'AOE', 'FPC', 'hclust', or 'alphabet'
rect.col	colour for the rectangular border
tl.cex	size of the text label
t1.col	colour of the text label
tl.srt	numeric value for text label string rotation

Another plotting function for the correlation matrix is the *ggcorrplot()* function, as illustrated below:

```
> install.packages("ggcorrplot")
```

```
Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-library/4.1'
```

```
...
** testing if installed package can be loaded from final location
```

```
** testing if installed package keeps a record of temporary installation path
```

```
* DONE (ggcorrplot)
```

```
> library("ggcorrplot")
```

```
Loading required package: ggplot2
```